



SUJAN BHARADWAJ

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Website: [Sujan Bharadwaj Portfolio](#), Visa Status: Visa Subclass 485 – Full-time Work Rights

SUMMARY

AI & Data Science Engineer with ~3 years of experience in machine learning, conversational AI, and cloud analytics. Proven track record in building knowledge graph-powered chatbots, ESG analytics platforms, and deploying ML models across sustainability, healthcare, and research domains. Proficient in Python, SQL, SPARQL, PyTorch, LLMs, RAG systems, AWS, and modern data visualization tools. Specialized in developing intelligent conversational systems that leverage knowledge graphs for enhanced searchability and user understanding. Adept at transforming complex data into production-ready AI solutions that generate measurable business impact through automation and AI-powered insights.

TECHNICAL SKILLS

- **Languages:** Python, C, SQL, SPARQL
- **ML & AI:** PyTorch, TensorFlow, Pandas, Scikit-learn, Transformers, LangChain
- **Knowledge Graphs:** RDFLib, Neo4j, SPARQL, Ontology Engineering
- **LLMs & NLP:** OpenAI API, Hugging Face, RAG, Vector Databases, Semantic Search
- **Cloud & Tools:** AWS, Azure, Docker, REST APIs, Git, Linux, Vector Stores
- **Visualization & BI:** Google Data Studio, QlikView, Matplotlib, Seaborn, Streamlit
- **Other:** Data Structures, System Design, OOP, Agile, GitHub, JIRA, Chatbot Development

EDUCATION

Master of Information Technology (Artificial Intelligence)
University of New South Wales (UNSW), Sydney, Australia

May'23 – May'25

Bachelor of Engineering - Electronics and Communication.
B.N.M Institute of Technology, Bengaluru, India.

Aug'16 - Aug '20

CERTIFICATIONS & PUBLICATIONS

- Microsoft Azure Fundamentals (**AZ-900**) Certified
- IRJET Publication: "**Multi-Account Embedded ATM Card**" (Embedded System Security) [Read Paper](#)

PROFESSIONAL EXPERIENCE

AI Engineer Intern

Apr'25 – Present

Scinter, Sydney, Australia

- **Designed and enhanced Large Concept Models (LCM)** to extract interdisciplinary insights from scientific literature and academic publications.
- **Built Dynamic Knowledge Graphs** mapping publications, entities, research trends, and semantic relationships for improved data discoverability.
- **Developing an intelligent chatbot system integrating Retrieval-Augmented Generation (RAG)** with knowledge graphs to enable natural language querying of scientific data and enhanced user searchability.
- **Implemented LLM-to-Knowledge Graph integration** using vector embeddings and semantic search to bridge conversational AI with structured knowledge representation, improving query accuracy and contextual understanding.
- **Structured unstructured scientific data into ontologies** for semantic search, concept reasoning, and interactive visual analytics.
- **Supported deployment of scalable NLP pipelines** for automated concept linking, trend forecasting, and knowledge graph population.



AI & Data Science Engineer (Part-Time)

May'23 – May'25

UNSW, Sydney, Australia

- Led end-to-end development of an **ontology-enhanced ESG analytics platform**, integrating domain-specific knowledge graphs, RDF ontologies, and SPARQL queries with principal component analysis (PCA) to produce interpretable sustainability insights across sectors.
- Built a reusable **ESG Knowledge Graph Engine** using Python and RDFLib, enabling automated feature selection, disclosure-topic classification, and industry-specific ESG profiling based on SASB standards.
- Designed and deployed **deep learning pipelines** for environmental and marine image segmentation:
 - Developed a U-Net model with Group Normalization and dropout layers to process RUGD terrain data under challenging lighting, achieving an 18% boost in pixel-level accuracy.
 - Applied CNN architectures to the SeaTurtleID2022 dataset, reaching 0.84 IoU and over 92% classification accuracy for marine species recognition.
- Created a **multi-label classification system** for disease detection in healthcare datasets using SMOTE for imbalance handling and batch normalization for improved convergence; achieved an average ROC-AUC > 0.91 across multiple conditions.
- Conducted **comparative benchmarking** of traditional and DL models (SVM, HOG+SVM vs CNN) across domain-shifted datasets, with deep learning models outperforming by 20–25% in F1-score.
- Documented all experiments in reproducible modules for potential open-source publication, emphasizing engineering standards, modularity, and cross-domain reusability.

Cloud Analytics Analyst:

Jan' 21 – Feb'23

Accenture, India

- **Automated end-to-end cloud cost analytics workflows** using Python, SQL, and REST APIs, reducing manual reporting time by **~40%** and enabling real-time anomaly detection for infrastructure cost spikes across multi-cloud environments (AWS, Azure).
- **Designed and deployed interactive KPI dashboards** in Google Data Studio and QlikView, enabling 20+ stakeholders across engineering and operations to track resource consumption, forecast trends, and make data-driven budget decisions.
- **Led a resource optimization initiative** that identified underutilized and idle cloud assets including EC2 instances, storage buckets, and Elastic IPs—resulting in **monthly cost savings of 25–30%** for enterprise clients.
- **Developed a suite of modular scripts and monitoring tools** to track usage patterns across AWS EC2, Azure Blob Storage, and RDS instances, improving visibility into resource consumption and streamlining compliance reporting for ISO and internal audits.
- **Bridged communication gaps between technical and financial teams** by aligning usage metrics with billing forecasts. Created standardized usage-to-cost mapping logic that improved billing accuracy by **15%**.
- **Mentored three junior analysts**, sharing best practices in cloud optimization and analytics scripting, and contributed reusable code modules to internal automation repositories—enhancing team productivity and onboarding efficiency.

IoT Intern – Health Monitoring System

July '19 – Aug'19

Nettur Technical Training Foundation

- **Engineered a real-time health monitoring system** integrating temperature, heart rate, and motion sensors with microcontrollers to continuously track patient vitals and trigger alerts for anomalies.
- **Programmed embedded systems** using C to handle real-time signal acquisition and processing, achieving response latency under 250ms, crucial for timely health interventions.
- **Implemented signal filtering and validation algorithms** to minimize false positives and ensure >95% system reliability in noisy test environments.

REFEREES

Contact Details Provided upon Request.